



First artificial human prion made

Researchers have created the first artificial human prion which may offer treatment for brain-wasting diseases. <https://digit.in/jul18-74-1>



Days getting longer

A new research indicates that days on Earth are getting longer as the Moon moves away from us. <https://digit.in/jul18-74-2>

SELF ASSEMBLING SPACE TELESCOPES

The coming generations of space telescopes are expected to be too large to fit in a single payload, and will thus require some kind of autonomous assembly in orbit.

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he primary mirror of the James Webb Space Telescope (JWST) is seen in the image above, with a

human on the side for scale. The mirror is 6.5 meters in diameter, an improvement over the 2.4 meter mirror of the Hubble Space Telescope. The JWST is designed to be powerful enough to resolve a single bumblebee on the face of the Moon. The JWST is scheduled for launch in 2020. It will be put into orbit as a single payload, with the components folding up. Towards the left side of the image, the gap between the hexagonal components, shows one of these folds. When launched, the JWST would be the largest astronomical instrument in space. However, engineers around the world are already coming up with ways to peer deeper into the universe, and at better resolutions.

The major component of a space telescope is a primary mirror. The size of the mirror is known as the aperture of the telescope. In the larger



Credit: NASA/Chris Gunn

space telescopes, this is invariably made up of multiple mirrors assembled together. The larger the primary mirror is, the better the telescope is. The parts of the mirror for future telescopes will have to be sent up to space over multiple launches. The materials for the mirror would have to be lightweight, to keep the cost of the launches low. The modular design should allow for easy assembly in space, and a mechanism for locking the various parts together. The components would require some kind of propulsion mechanism to arrive at the rendezvous point, after the launch. Then there has to be a way to assemble these components together in space. The parts of the mirror can navigate to each other and assemble themselves. Another option is that humans or robots in space can intervene during the assembly process. In all, it is preferable if the

entire assembly process can be automated. There is also a requirement of deploying a Sun shield of sorts. This is a preferably thin screen that can protect the components from solar activity, and keep the temperature of the instruments low, which are ideal for the health of the telescopes.

NASA has just approved a feasibility study for one such telescope. The proposal is that of a modular active self-assembling space telescope swarm. Dmitry Savransky from Cornell University and 15 other engineers have been developing a concept for a telescope that could autonomously assemble in space, and require a number of relatively cheap launches involving small payloads in each flight. The idea is to use a number of hexagonal modules, each 1 meter in diameter. Every module is autonomous, and is loaded with a number of subsystems such as actuators,